Calcurion Maths Worksheets

GCSE Mathematics

Topic: Sequences

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Find the next two terms of the sequence $3, 7, 11, 15, \ldots$. **[2]**

2. Find the next two terms of the sequence $20, 17, 14, 11, \ldots$. **[2]**

3. Find the next two terms of the sequence $2, 6, 18, 54, \ldots$. **[2]**

4. Find the nth term of the sequence $5, 8, 11, 14, \ldots$. **[3]**

5. Find the nth term of the sequence $4, 7, 10, 13, \ldots$. **[3]**

6. Find the nth term of the sequence $3, 7, 11, 15, \ldots$. **[3]**

7. Find the nth term of the sequence $20, 17, 14, 11, \ldots$. **[3]**

8. Is 50 a term of the sequence $5, 8, 11, 14, \ldots$? Justify your answer. **[3]**

9. Find the 20th term of the sequence $2, 5, 8, 11, \ldots$. **[3]**

10. Find the 15th term of the sequence $100, 95, 90, 85, \ldots$. **[3]**

11. The nth term of a sequence is $3n + 2$. Find the first three terms. **[2]**

12. The nth term of a sequence is $5n - 1$. Find the 10th term. **[2]**

13. Find the next term of the sequence $1, 4, 9, 16, 25, \ldots$. **[2]**

14. Find the next term of the sequence $1, 8, 27, 64, \ldots$. **[2]**

15. Find the next two terms of the Fibonacci sequence $1, 1, 2, 3, 5, 8, \ldots$. **[2]**

Section B – Hard (Q16–25)

16. Find the nth term of the sequence $2, 8, 18, 32, 50, \ldots$. **[4]**

17. Find the nth term of the sequence $3, 12, 27, 48, 75, \ldots$. **[4]**

18. The nth term of a sequence is $n^2 + 3$. Find the 8th term. **[2]**

19. The nth term of a sequence is $2n^2 - 1$. Is 49 a term of the sequence? **[4]**

20. Find the nth term of the sequence $5, 11, 21, 35, 53, \ldots$. **[5]**

21. An arithmetic sequence has first term 7 and common difference 4. Find the 25th term. **[3]**

22. An arithmetic sequence has first term a and common difference d. The 5th term is 19 and the 10th term is 34. Find a and d. **[5]**

23. Find the sum of the first 10 terms of the sequence $3, 7, 11, 15, \ldots$. **[4]**

24. A geometric sequence has first term 5 and common ratio 2. Find the 6th term. **[3]**

25. The first term of a geometric sequence is 3 and the common ratio is $\frac{1}{2}$. Find the 5th term. **[3]**

End of Worksheet • *Show all working clearly.*